

Module 01: Systems — The Robot Factory

Learning Objectives

1. Understand that a **Robot** is a System made of many parts.
2. Learn the 3 main parts: **Sensors** (Eyes), **Computer** (Brain), and **Actuators** (Muscles).
3. Know that the parts must work together or the robot won't work.

Introduction: The Robot Factory

Welcome to the Robot Factory! We are building a machine helper. Is a robot just a piece of metal? No! It is a **System**.

- It needs **Batters** to have energy.
- It needs **Wires** to send messages.
- It needs **Code** to know what to do.

If you take out the battery, does the robot work? No! The System is broken. All parts must work as a team.

Key Concepts

1. Inputs (Sensors)

Robots need to see the world. They use **Sensors** (Cameras, Microphones, Bumpers). This is like your Eyes and Ears. It brings information **IN**.

2. Processing (The Computer)

Inside the robot is a computer chip. It thinks: "I see a wall. I should stop." This is the **Controller**.

3. Outputs (Actuators)

Robots need to move. They use **Motors** (Wheels, Arms, Grippers). This sends action **OUT**.

Activities

Activity 1: Be the Robot

Play with a partner.

- **Part 1:** Use your "Sensors" (Close your eyes. Partner taps your shoulder).
- **Part 2:** Use your "Computer" (Think: A tap means Turn Left).
- **Part 3:** Use your "Actuators" (Turn your body left). You are a human system acting like a robot!

Activity 2: Design a Bot

Draw a robot on paper. Label its parts:

- Where are the Sensors? (Eyes?)
- Where is the Computer? (Head?)
- Where are the Actuators? (Wheels? Legs?)
- Where is the Power? (Battery pack?)

Summary

A robot is a special system that can Sense, Think, and Act. It is efficient because all its parts (Electronic and Mechanical) work together perfectly!

References

- *Boy and Bot* by Ame Dyckman
- *The Wild Robot* by Peter Brown (For older kids)